SE lab exam

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SY B.tech CE
Batch A

Github link:

	A	B	C	D	E	F
1	Test Case ID	Scenario	Input	Expected Output	Actual Output	Pass/Fail
2	TC01	Valid delivery date	Booking:10 May, Delivery:12 May	Accepted		pass
3	TC02	Delivery = booking date	Booking:10 May, Delivery:10 May	Error		pass
4	TC03	Delivery before booking	Booking:10 May, Delivery:9 May	Error		pass
5	TC04	Just after booking	Booking:10 May, Delivery:11 May	Accepted		pass
6	TC05	Far future date	Booking:10 May, Delivery:30 May	Accepted		pass
7	TC06	Null delivery date	Delivery: NULL	Error		pass
8	TC07	Invalid date format	Delivery: abc	Error		pass
9	TC08	Package within limit	Count = 3	Accepted		pass
10	TC09	Package at max limit	Count = 5	Accepted		pass
11	TC10	Package exceeds limit	Count = 6	Error		pass
12	TC11	Just below max	Count = 4	Accepted		pass
13	TC12	Zero package	Count = 0	Error		pass
14	TC13	Negative package	Count = -1	Error		pass
15	TC14	Null package count	Count = NULL	Error		pass
16	TC15	Non-numeric package	Count = five	Error		pass
17						
18						
19						

```
● Yash(221080057) \w $ g++ -std=c++11 test_black_box.cpp BookingSystem.cpp Shipment.cpp Sender.cpp -o test_black_box && ./test_black_box
  ================================================================================
  ID    Scenario                  Expected Output                                  Result
  ================================================================================
  TC01  Valid delivery date       Accepted                                         PASS
  TC02  Delivery = booking date   Error: Delivery date must be after booking dat   PASS
  TC03  Delivery before booking   Error: Delivery date must be after booking dat   PASS
  TC04  Just after booking        Accepted                                         PASS
  TC05  Far future date           Accepted                                         PASS
  TC06  Null delivery date        Error: Invalid input (null/negative/non-numeri   PASS
  TC07  Invalid date format       Error: Invalid input (null/negative/non-numeri   PASS
  TC08  Package within limit      Accepted                                         PASS
  TC09  Package at max limit      Accepted                                         PASS
  TC10  Package exceeds limit     Error: Package limit exceeded (Max = 5)          PASS
  TC11  Just below max            Accepted                                         PASS
  TC12  Zero package              Error: Invalid input (null/negative/non-numeri   PASS
  TC13  Negative package          Error: Invalid input (null/negative/non-numeri   PASS
  TC14  Null package count        Error: Invalid input (null/negative/non-numeri   PASS
  TC15  Non-numeric package       Error: Invalid input (null/negative/non-numeri   PASS
  ================================================================================
  Total Black Box Tests Passed: 15 / 15
```

	A	B	C	D	E	F	G
1	TC ID	currentStatus	newStatus	daysInTransit	isPaymentCleared	Expected Output	
2	TC01	Pending	Dispatched	0	FALSE	Error: Payment not cleared	Pass
3	TC02	Pending	Pending	0	TRUE	Error: Status already Pending	Pass
4	TC03	Pending	Dispatched	0	TRUE	Success: Dispatched	Pass
5	TC04	Pending	Cancelled	0	TRUE	Success: Cancelled	Pass
6	TC05	Pending	Delivered	0	TRUE	Error: Invalid transition from Pending	Pass
7	TC06	Dispatched	In Transit	1	TRUE	Success: In Transit	Pass
8	TC07	Dispatched	In Transit	0	TRUE	Error: Invalid transition from Dispatched	Pass
9	TC08	Dispatched	Delivered	5	TRUE	Error: Cannot skip In Transit	Pass
10	TC09	Dispatched	Cancelled	0	TRUE	Error: Invalid transition from Dispatched	Pass
11	TC10	In Transit	Delivered	2	TRUE	Success: Delivered	Pass
12	TC11	In Transit	Delivered	1	TRUE	Error: Invalid transition from In Transit	Pass
13	TC12	In Transit	Lost	8	TRUE	Status Updated to Lost	Pass
14	TC13	In Transit	Lost	7	TRUE	Error: Invalid transition from In Transit	Pass
15	TC14	Delivered	Return	0	TRUE	Error: Terminal state	Pass
16	TC15	Unknown	Pending	0	TRUE	Error: Unknown Status	Pass

```
Yash(221080057) \w $ g++ -std=c++11 test_white_box.cpp Shipment.cpp -o test_white_box && ./test_white_box
======================================================================
ID    Expected                      Actual                        Result
======================================================================
TC01  Error: Payment not cleared    Error: Payment not cleared    PASS
TC02  Error: Status already Pendin  Error: Status already Pendin  PASS
TC03  Success: Dispatched           Success: Dispatched           PASS
TC04  Success: Cancelled            Success: Cancelled            PASS
TC05  Error: Invalid transition fr  Error: Invalid transition fr  PASS
TC06  Success: In Transit           Success: In Transit           PASS
TC07  Error: Invalid transition fr  Error: Invalid transition fr  PASS
TC08  Error: Cannot skip In Transi  Error: Cannot skip In Transi  PASS
TC09  Error: Invalid transition fr  Error: Invalid transition fr  PASS
TC10  Success: Delivered            Success: Delivered            PASS
TC11  Error: Invalid transition fr  Error: Invalid transition fr  PASS
TC12  Status Updated to Lost        Status Updated to Lost        PASS
TC13  Error: Invalid transition fr  Error: Invalid transition fr  PASS
TC14  Error: Terminal state         Error: Terminal state         PASS
TC15  Error: Unknown Status         Error: Unknown Status         PASS
======================================================================
Total Passed: 15 / 15
Yash(221080057) \w $
```

Sample Output for main.cpp

g++ -std=c++11 main.cpp BookingSystem.cpp Shipment.cpp Sender.cpp -o booking_system && ./booking_system

```
Yash(221080057) \w $ g++ -std=c++11 main.cpp BookingSystem.cpp Shipment.cpp Sender.cpp -o booking_system && ./booking_system

Enter Sender Name: Soham
Enter Receiver Name: Rahul
Enter Address: Mumbai
Enter Package Weight: 3
Enter Service Type: express
Enter Booking Date: 10-05-2026
Enter Delivery Date: 15-05-2026
Enter Package Count: 3

Booking Successful!
Tracking ID: TRK12345
Service Type: express
Shipment Label Generated
Yash(221080057) \w $
```